MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-12044	- Chest, Plywood.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13218E6736	- Tester, Leak Detector, for Seams in Liquid Storage Tanks.
D13218E6764	- Chest, Vacuum Seam Tester.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 North West Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The vacuum seam tester shall be in accordance with Top Assembly TA13218E6736 and as specified herein.

3.1.1 Chest. The chest shall be constructed in accordance with Drawing No. D13218E6764 and MIL-C-12044.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the vacuum seam tester. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a

ENCY'S SUPPLEMENTAL THE DETERMINATION OF SMALL BUSINESS STATUS, NUMBER OF LIST OR IN THE FEDERAL RE
Source: https://assist.dla.mil -- Downloaded: 2026-01-18T03:17Z
Check the source to verify that this is the current version before use.

vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced vacuum seam tester and chest). The contractor shall furnish one vacuum seam tester with chest for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce vacuum seam testers and chests that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the vacuum seam tester shall conform to MIL-STD-461, Class IIB.

3.5 Performance. The vacuum seam tester shall be tested as specified in 4.5.2.2.

3.6 Identification marking. The vacuum seam tester and chest shall be identified in accordance with MIL-STD-130. The marking shall be applied to the tester on plates conforming to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.7 Instruction plates. Each vacuum seam tester shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the seam tester. Instruction plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.8 Treatment and painting. The portions of the vacuum seam tester and chest normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

3.8.1 Stenciling. The nomenclature and identification marking shall be stenciled on the base of the vacuum seam tester. Lettering shall be not less than 1/2-inch (13 mm) high where space permits.

3.9 Workmanship. The quality of workmanship shall be such as to produce vacuum seam testers and chests that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the units.

3.9.1 Castings. All parts, components, and assemblies of the vacuum seam tester which include castings shall be cleaned of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.9.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the vacuum seam tester to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch (4.762 mm) unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the vacuum seam tester shall be at the option of the contractor.

3.9.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.9.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.9.5 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.

(b) Tack welds shall be subject to the same quality requirements as final welds.

(c) Work shall be positioned for flat welding whenever practicable.

(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and vacuum seam tester.

3.9.6 Weld quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the vacuum seam tester, the vacuum seam tester shall conform to the inspection requirements specified in Section 4.

(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.

(c) Undercut in weldments shall be held to a minimum and shall not be more than 1/32-inch (0.794 mm) deep.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

3.9.7 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.9.8 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced vacuum seam tester and chest shall be cause for disassembly, in the presence of a Government representative, of the first produced vacuum seam tester and chest to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced vacuum seam tester and chest. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

(a) First produced item inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced inspection.

4.3.1 Examination. The first produced vacuum seam tester and chest shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced vacuum seam tester shall be tested as specified in 4.5.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of the tests shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The vacuum seam leak detector and chest shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the requirements shown on the drawings.
102. Material not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

103. Components missing or not as specified.
104. Treatment and painting not as specified.
105. Workmanship not as specified.

Minor

201. Identification marking illegible or missing.
202. Stenciling incorrect or missing.

4.5.2 Tests.

4.5.2.1 Electromagnetic interference control. The vacuum seam tester shall be tested in accordance with MIL-STD-462 to determine compliance with the requirements of MIL-STD-461. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.2 Performance. The vacuum seam tester shall be tested by operating the unit for not less than 5 minutes with the inspection box on a smooth surface. The tester should develop 3 pounds per square inch (psi) vacuum in not more than 10 seconds and maintain the vacuum for an additional 5 minutes. The vacuum box should hold a vacuum of 10 psi without loss for a period of not less than 5 minutes with the valve closed. Inability to operate as specified shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Materials, methods, or containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:17Z
Check the source to verify that this is the current version before use.

107. Unprotected surfaces not coated with preservative as specified for Level A.
108. Vacuum pump not preserved and opening sealed as specified for Level A.
109. Consolidation not as specified for Level A.
110. Packing not as specified for Level A or Level B.
111. Strapping not as specified for Level A.
112. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unprotected exterior metal surfaces of components requiring application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Vacuum pump. Interior surfaces of the pump shall be coated with preservative lubricating oil conforming to MIL-L-21260, Type I, Grade 30, and the excess oil drained. Openings into the pump shall be sealed with tape conforming to PPP-T-60, Type IV.

5.1.1.3 Wedge gaskets. Wedge gaskets shall be preserved together in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.4 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.5 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.1.1.6 Consolidation. The components comprising a complete tester shall be consolidated in the chest as shown on the drawings. Cushioning conforming to PPP-C-843, Type II, Class A, shall be provided, as required, to prevent movement of contents. The cover shall be closed and secured.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:17Z
Check the source to verify that this is the current version before use.

5.1.2 Commercial. Testers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Each chest containing a complete tester, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

5.2.2 Level B. Each chest containing a complete tester, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional. Box closure and reinforcement requirements shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Each chest containing a complete tester, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The vacuum seam tester is intended for use in testing the seams in the floor of welded or bolted tanks for leaks prior to being put into service.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced vacuum seam tester and chest (see 3.2).
(c) Welding process to be used (see 3.9.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

6.3 First produced vacuum seam tester and chest. Any changes or deviations of production vacuum seam testers and chests from the approved first produced vacuum seam tester and chest during production will be subject to the approval of the contracting officer. Approval of the first produced vacuum seam tester and chest will not relieve the contractor of his obligation to furnish vacuum seam testers and chests conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each vacuum seam tester and chest.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of vacuum seam testers and chests.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 5430-0123

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-14419B(ME) Tester, Leak Detector, Vacuum Type for Seams in Bolted and Welded Liquid Storage Tanks

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:17Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:17Z
Check the source to verify that this is the current version before use.